Software Requirements Specification

Hasnahena Web Application

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Interest-free charitable community lending platform based on Islamic financial principles

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Contents

Document Control **2**

1 Introduction **3**
1.1 Purpose . . . 3
1.2 Project Scope . . . 3
1.3 Definitions & Acronyms . . . 3

2 Overall Description **3**
2.1 Product Perspective . . . 3
2.2 User Roles & Permissions . . . 3
2.2.1 Member . . . 3
2.2.2 Worker (Staff) . . . 4
2.2.3 Admin . . . 4

3 Functional Requirements **4**
3.1 Authentication & Access Control . . . 4
3.2 Member Registration . . . 4
3.3 Financial Transactions . . . 5
3.4 Loan Management . . . 5
3.5 Donations . . . 5
3.6 Dispute & Audit . . . 6

4 Non-Functional Requirements **6**
4.1 Security . . . 6
4.2 Compliance . . . 6
4.3 Usability . . . 6
4.4 Performance . . . 6
4.5 Scalability . . . 6

5 Appendices **7**
5.1 Requirement Traceability . . . 7
5.2 Approvals . . . 7

Document Control

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Revision History

Ver.	Date	Author	Changes
1.0	—	Md Saif Ahmed	Initial draft
2.0	—	Md Saif Ahmed	Added guarantor mechanism
3.0	March 22, 2026	Md Saif Ahmed	Final release with all enhancements

1 Introduction

1.1 Purpose

This document serves as the Software Requirements Specification (SRS) for Hasnahena, an interest-free, charitable community lending platform based on Islamic financial principles (*Qard al-Hasan*).

The SRS defines the complete set of functional and non-functional requirements that the system must satisfy to meet stakeholder needs.

1.2 Project Scope

The Hasnahena web application provides a secure portal for community members to:

- Deposit savings without earning interest
- Borrow funds without paying interest
- Act as guarantors for other members
- Make charitable donations (Zakat/Sadaqah)

Roles are uniquely designated into **Members**, **Workers**, and **Admins** to ensure strict financial integrity and complete access logging.

1.3 Definitions & Acronyms

Term	Definition
BDT	Bangladeshi Taka (currency)
NID	National Identity Document
RBAC	Role-Based Access Control
Qard al-Hasan	Interest-free loan in Islamic finance
Zakat	Obligatory charitable giving in Islam
Sadaqah	Voluntary charitable giving in Islam

2 Overall Description

2.1 Product Perspective

The web application is built on a modern microservices architecture:

- **Frontend:** Responsive Next.js (React) application with TypeScript
- **API Gateway:** Go-based edge service for traffic handling
- **Core Ledger:** Java Spring Boot service for double-entry bookkeeping
- **Database:** PostgreSQL with ACID transaction guarantees

2.2 User Roles & Permissions

2.2.1 Member

The primary user of the system.

- Can deposit, borrow (after meeting minimum criteria), act as guarantor, and donate

- All financial flows are **initiated by members**
- Can raise disputes for incorrect transactions
- Can update nominee information

2.2.2 Worker (Staff)

Internal staff responsible for fund processing.

- Processes digital/physical fund transfers based on member requests
- **Cannot** delete records or initiate deposits without a request
- Records loan repayment installments
- Can record offline donations on behalf of members

2.2.3 Admin

Highest privilege role for system oversight.

- Manages worker accounts and permissions
- Approves/rejects member registration and loan requests
- Reverses erroneous transactions (with audit logging)
- Configures system parameters (loan ratios, fees, minimums)

3 Functional Requirements

3.1 Authentication & Access Control

REQ-AUTH-01: The system shall support login using phone number or email with password
Priority: High

REQ-AUTH-02: The system shall enforce mandatory password change on first login after approval
Priority: High

REQ-AUTH-03: The system shall implement RBAC for all protected endpoints
Priority: Critical

3.2 Member Registration

REQ-REG-01: The system shall allow members to create accounts with NID and nominee details
Priority: High

REQ-REG-02: The system shall queue new registrations for admin approval
Priority: High

REQ-REG-03: The system shall send notification (SMS/email) upon approval or rejection
Priority: Medium

3.3 Financial Transactions

REQ-FIN-01: The system shall require a minimum deposit of 8,000 BDT for loan eligibility
Priority: High

REQ-FIN-02: Workers shall process financial transactions pending final confirmation from members
Priority: Critical

REQ-FIN-03: The system shall enforce 3-month notice period for withdrawals
Priority: High

REQ-FIN-04: The system shall block withdrawals that violate loan collateral ratio
Priority: Critical

3.4 Loan Management

REQ-LOAN-01: The system shall calculate maximum loan based on personal cap (5× deposit), fairness cap, and solvency cap
Priority: High

REQ-LOAN-02: The system shall support guarantor request and approval workflow
Priority: High

REQ-LOAN-03: The system shall execute default settlement waterfall (borrower → guarantor → charity fund)
Priority: Critical

3.5 Donations

REQ-DON-01: The system shall categorize donations into Zakat, Sadaqah, and Operational Funds
Priority: High

REQ-DON-02: Donation funds shall remain isolated from withdrawable member deposits
Priority: Critical

3.6 Dispute & Audit

REQ-AUD-01: The system shall provide dispute tracking if a worker processes an incorrect amount

Priority: High

REQ-AUD-02: The system shall maintain an append-only, immutable audit log for all financial actions

Priority: Critical

4 Non-Functional Requirements

4.1 Security

- Enforce Role-Based Access Control (RBAC) on all protected resources
- Database transaction atomicity for all financial operations
- Support for multi-factor authentication (future phase)
- Rate limiting on authentication and sensitive endpoints
- All sensitive data encrypted at rest and in transit

4.2 Compliance

- Fully Shariah-compliant (zero interest, full transparency)
- Robust auditing trails for regulatory compliance
- Data retention policies compliant with local regulations

4.3 Usability

- Full bilingual support (English/Bengali)
- Responsive design for desktop and mobile browsers
- Accessibility compliance (WCAG 2.1 Level AA target)

4.4 Performance

- Page load time under 3 seconds on standard connections
- API response time under 500ms for 95th percentile
- System availability target: 99.5%

4.5 Scalability

- Support for up to 10,000 concurrent members in Phase 1
- Horizontal scaling capability for backend services

5 Appendices

5.1 Requirement Traceability

All requirements (REQ-*) are traceable to:

- Business Requirements Document (BRD)
- Use Case Specifications
- Test Plan and Strategy

5.2 Approvals

Technical Lead	**Product Owner**
Date: _____	Date: _____